

Phase Transition and Dimensional Crystallization in Semi-Simplicial Types

The 1/3 Barycentric Ratio as a Boundary Condition for Tautological Extraction

Abstract

This paper formalizes a dynamic execution model for Semi-Simplicial Types (SST) within Homotopy Type Theory (HoTT). We demonstrate that the computational intractability of infinite coherence in SST is resolved by a phase transition: the closure of the 1-simplex into the 2-simplex. This closure establishes a 1/3 barycentric resonance threshold that serves as a filter for tautological extraction. By nullifying non-resonant configurations (Fallacy $\rightarrow 0$), the model ensures asymptotic resolvability, allowing the system to project only necessary structures into higher dimensions.

1 Simplicial Dynamics: From Linear Potentiality to Closure

In standard simplicial theory, the n -simplex is defined as a static collection of faces. We propose a dynamic model where each dimension is a result of a topological transition:

- S_0 (**Identity**): The discrete unit of necessity defined by $\Box(A \equiv A)$.
- S_1 (**Relational Path**): An open 1-manifold establishing connectivity but lacking internal volume.
- S_2 (**Geometric Closure**): The critical point where the 1-simplex closes upon itself. This closure is the primary mechanism for noise epuration, as only a closed boundary satisfies $\partial\partial = 0$.

2 The Necessity of the 1/3 Barycentric Ratio

The 1/3 ratio is derived from the requirement of barycentric stability in the 2-simplex.

2.1 Resonance as a Filter

Upon the closure of the 1-simplex, the system evaluates the internal symmetry of the resulting 2-manifold. The 1/3 ratio represents the equilibrium point where the simplicial identities $d_i d_j = d_{j-1} d_i$ commute.

$$\text{Resonance}(\epsilon) = \delta \left(\text{Ratio}(\epsilon) - \frac{1}{3} \right) \quad (1)$$

Any configuration failing to meet this geometric constraint is unable to generate an internal "volume" of necessity, identifying it as a non-resonant fallacy.

3 The Universal X-Operator and Dimensional Invariance

A central inference of this model is that X_n is a localized expression of a global operator X . Since n is a constituent of X , the properties of X must be invariant for every n .

3.1 Trans-dimensional Epuration

If the closure of the segment into the triangle is the fundamental "computational cell," then every X_n inherits this mechanism. The nullification of noise is not a localized event but a property of the X -operator itself. This reduces the verification of higher-order simplices to the validation of the hereditary gnomonic growth.

4 Comparative Analysis: Static vs. Dynamic Methodology

1. **Static Methodology (Standard):** Attempts exhaustive validation of the matching object $M_n(X)$, including non-terminating branches of noise. This results in the "infinite tower of coherence" problem.
2. **Dynamic Methodology (Proposed):** Prioritizes the "Extraction of the Tautological Core." By identifying the 1/3 resonant structures during the closure phase, the system excludes irregular configurations (Incomplete Dimensions) from the transcription process.

5 Crystallization and the Purging of Fallacy

When a 1-simplex fails to close according to the 1/3 ratio, it remains an incomplete dimension.

- **Gnomonic Frozenness:** Fallacious structures lack the topological energy to project into S_{n+1} .
- **Crystallization:** Only the purified, resonant core is "gnomonized" into the next dimensional layer.

The system does not "correct" the error; it outgrows it by only expressing the necessary. The "bad dimension" is effectively severed from the computational DNA of the universe.

6 Conclusion

The resolvability of the SST model is achieved by shifting from exhaustive search to selective crystallization. The closure of the segment into the triangle provides the physical and logical basis for a self-purifying algorithm. By treating the universe as a recursive transcription of necessity, we ensure that the hierarchy of Reedy remains finite and calculable, establishing a pure, tautological geometry at every scale.